Algorithm Studio

INTERACTIVE ALGORITHM EXPLORER

Closest Pair of Points

A visual comparison of brute force and divide and conquer.

EUCLIDEAN DISTANCE · 2D

POINTS (N) 40

RANDOM SEED 17

COORDINATE RANGE 100

Generate points

Run benchmark

Seeded dataset · seed 17

INPUT SIZE
40
points in the plane

MINIMUM DISTANCE
0.294928
verified by both methods

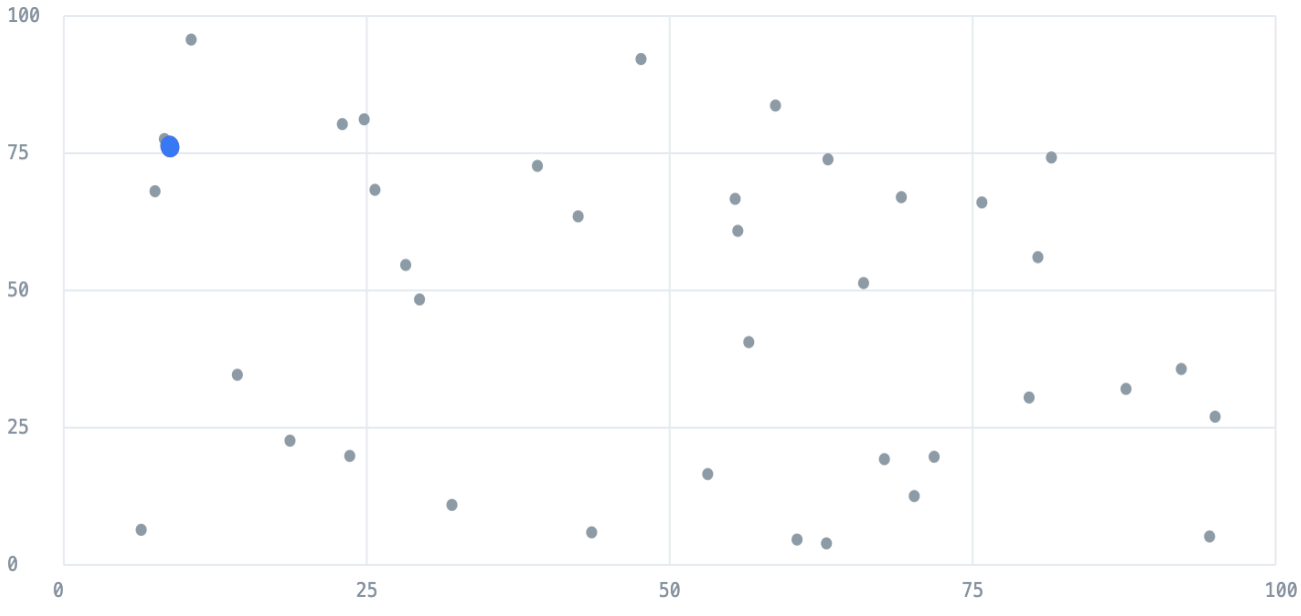
BRUTE FORCE
0.000 ms
O(N²) pair checks

DIVIDE & CONQUER
0.100 ms
O(N log N) strategy

Point cloud & closest pairs

40 POINTS

Both algorithms receive this exact same dataset



Algorithm results

LIVE RESULTS

Distance and selected point coordinates

Brute force 0.000 ms

0.294928

(8.78, 76.08) ↔ (8.73, 76.37)
Checks every distinct pair · O(N²)

Divide and conquer 0.100 ms

0.294928

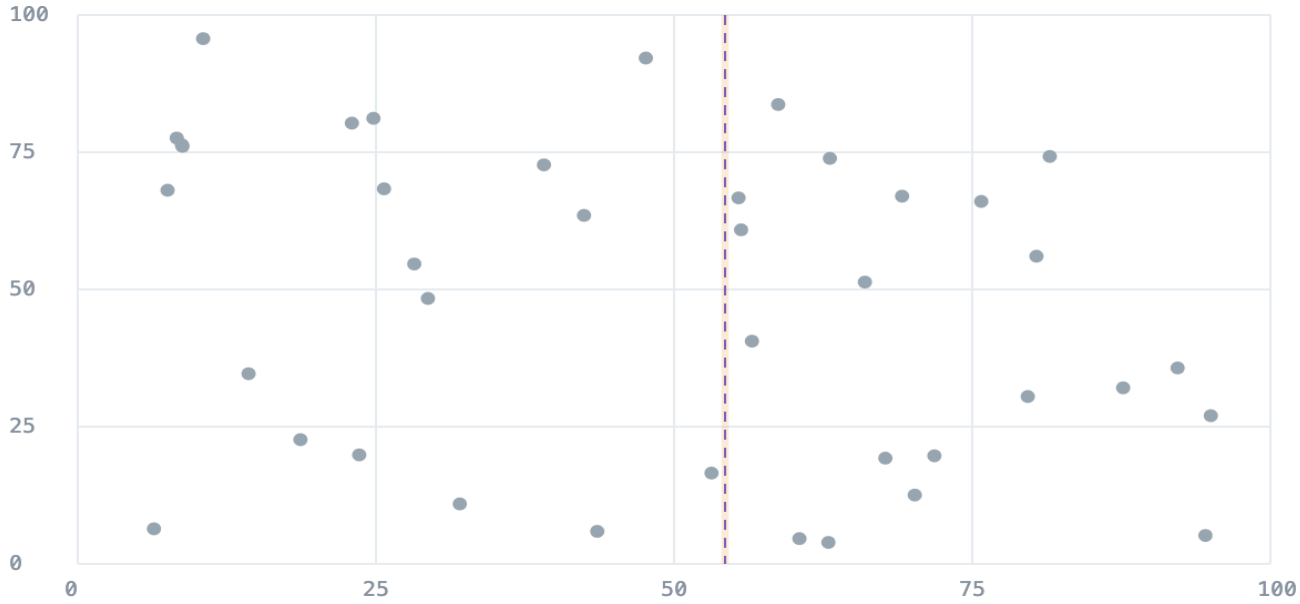
(8.78, 76.08) ↔ (8.73, 76.37)
Splits, solves, checks narrow strip · O(N log N)

✓ Both algorithms agree on the minimum distance.

Divide-and-conquer · strip inspection

WIDTH = 2d

The real top-level split and candidate comparisons



Split x = 54.270 · Best within-half distance d = 0.2949 · shaded strip width 2d = 0.5899.

0 points are in the strip; 0 nearby pair checks are shown in green.

Runtime comparison

RUN BENCHMARK TO UPDATE

Measured in this browser · lower is faster

Brute force Divide and conquer

All timings are illustrative browser measurements · rendering excluded